

Transactions & ACID Properties

LAB 1

```
30 SELECT * FROM rental
31 WHERE rental_id = 2;
```

Data Output Messages Notifications

Showing rows: 1 to 1 of 1

	rental_id [PK] integer	rental_date timestamp without time zone	inventory_id integer	customer_id smallint	return_date timestamp without time zone	staff_id smallint	last_update timestamp without time zone
1	2	2005-05-24 22:54:33	1525	459	2005-05-28 19:40:33	1	2006-02-16 02:30:53

```
30 --lab 1
31 SELECT * FROM rental
32 WHERE rental_id = 2;
33
34 BEGIN;
35     UPDATE rental SET rental_date = NOW() WHERE rental_id = 2;
36
37     INSERT INTO payment (customer_id, staff_id, rental_id, amount, payment_date)
38     VALUES (459, 1, 2, 99.99, 'WRONG_DATE_FORMAT');
39 COMMIT;
40
41
42
```

Data Output Messages Notifications

Showing rows: 1 to 1 of 1

ERROR: invalid input syntax for type timestamp: "WRONG_DATE_FORMAT"
LINE 5: VALUES (459, 1, 2, 99.99, 'WRONG_DATE_FORMAT');
^

SQL state: 22007
Character: 174

```
41 SELECT return_date FROM rental WHERE rental_id = 2;
42
```

Data Output Messages Notifications

Showing rows: 1 to 1 of 1

	return_date timestamp without time zone
1	2005-05-28 19:40:33

Successful

Is the return_date updated? Why not?

- No, the return_date is not updated.
- Due to the Atomicity property ("All or Nothing") of transactions. When the INSERT statement fails with an error, the entire transaction is rolled back, undoing the previous successful UPDATE statement so no partial data changes are saved

Exercise 2: The "Flexible" Staff Task

```
59
60 BEGIN;
61 INSERT INTO staff (first_name, last_name, address_id, email, store_id, username)
62 VALUES ('Alex', 'Staff', 1, 'alex.staff@email.com', 1, 'alex');
63
64 SAVEPOINT staff_saved;
65
66 UPDATE staff SET store_id = 999 WHERE username = 'alex';
67
68 ROLLBACK TO staff_saved;
69 UPDATE staff SET store_id = 1 WHERE username = 'alex';
70 COMMIT;
71
72
73
74
75
```

Data Output Messages Notifications

COMMIT

Query returned successfully in 77 msec.

```
72
73 SELECT * FROM staff;
74
75
```

Data Output Messages Notifications

Showing rows: 1 to 3 Page No: 1 of 1

	staff_id [PK] integer	first_name character varying (45)	last_name character varying (45)	address_id smallint	email character varying (50)	store_id smallint	active boolean	username character varying (16)	password character varying (40)
1	1	Mike	Hillyer	3	Mike.Hillyer@sakilastaff.com	1	true	Mike	8cb2
2	2	Jon	Stephens	4	Jon.Stephens@sakilastaff.c...	2	true	Jon	8cb2
3	3	Alex	Staff	1	alex.staff@email.com	1	true	alex	[null]